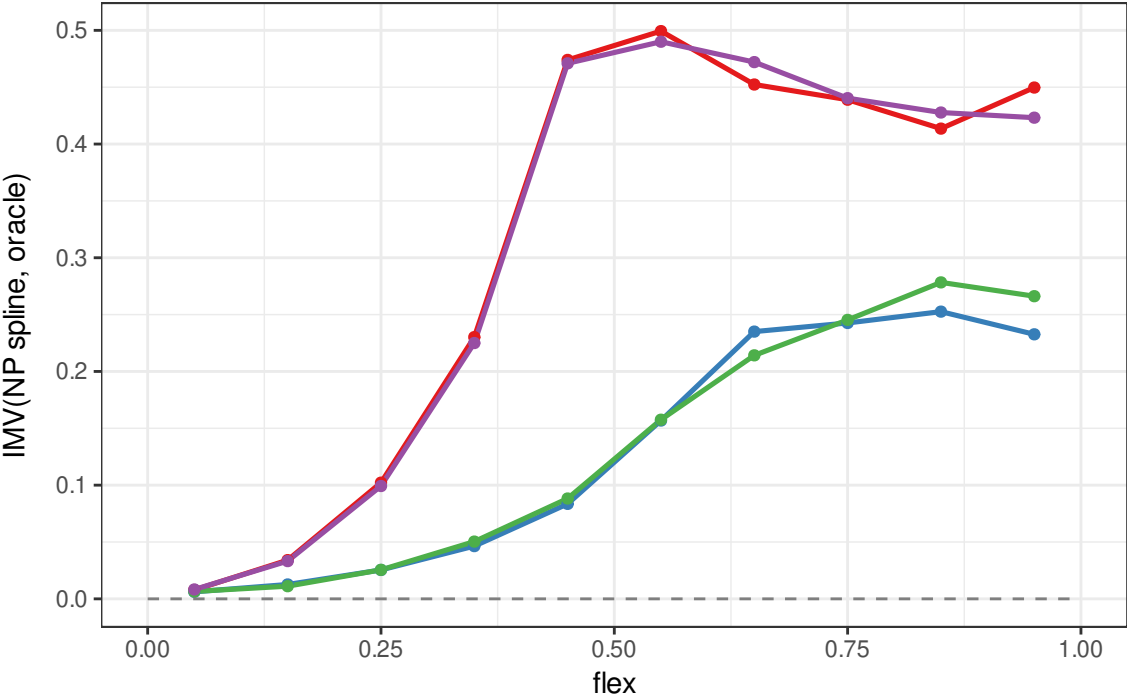


**$\omega_t$  decomposition: NP spline vs oracle by threshold and flex**

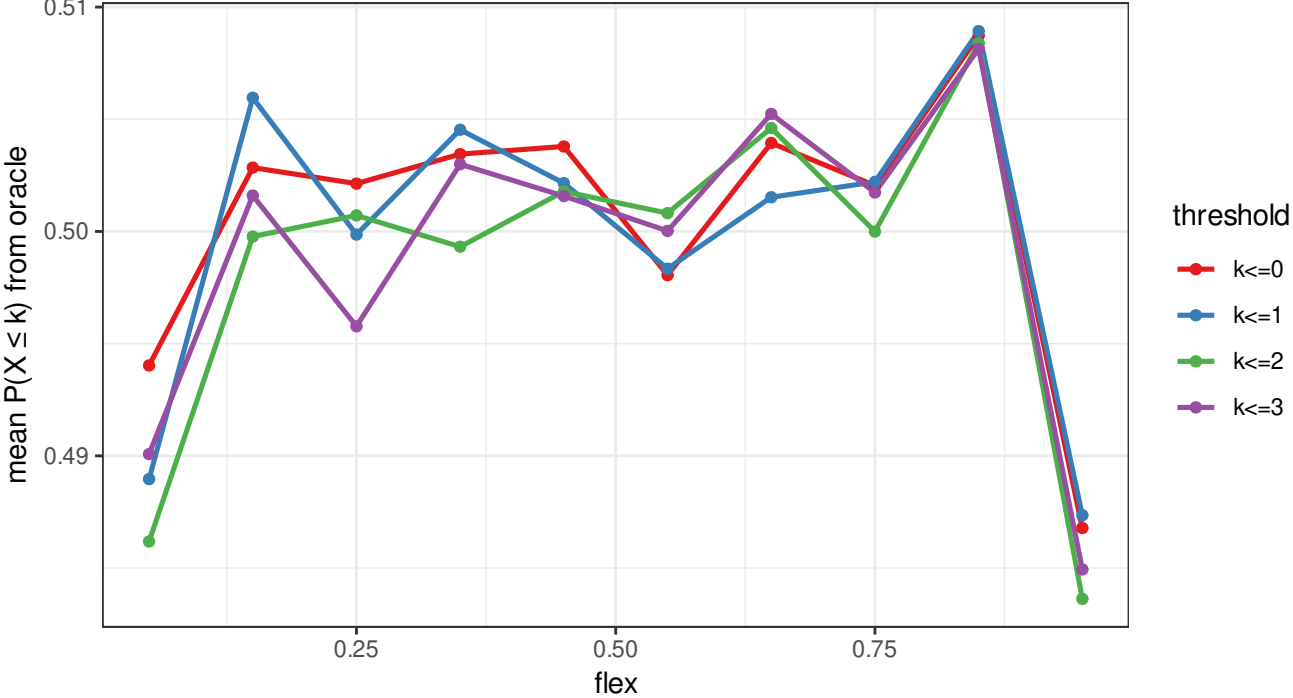
A: per-threshold binary IMV (oracle vs NP spline)

positive = oracle wins at that threshold

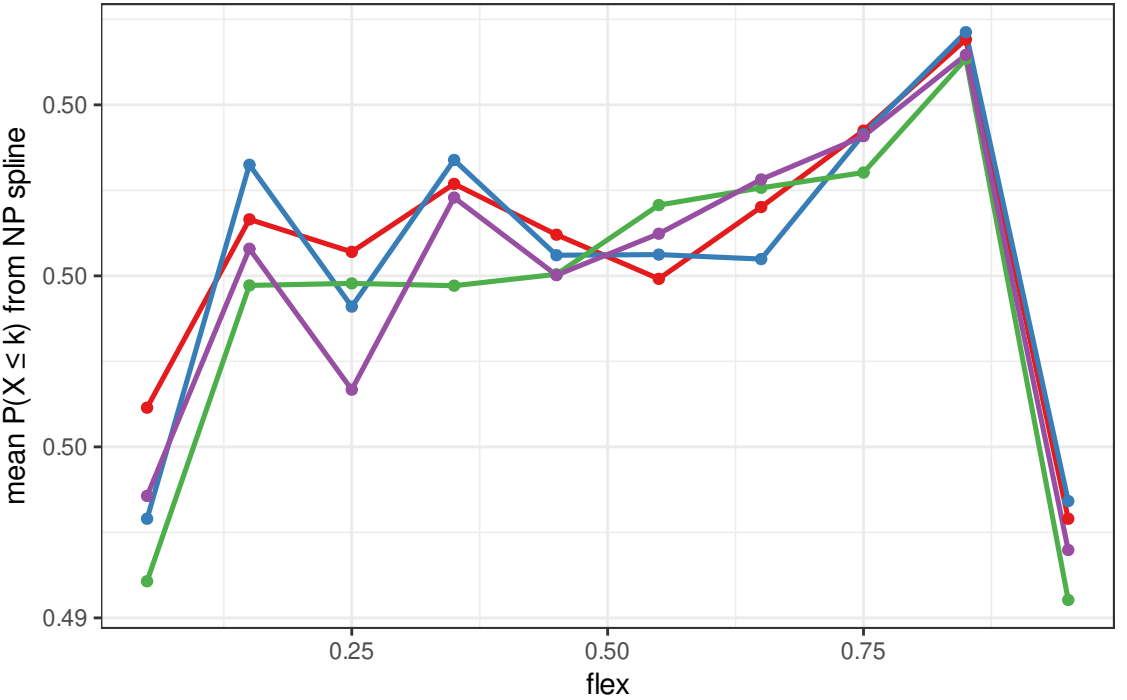


B: mean oracle cumulative probability  $P(X \leq k)$

approaches 0.5 as flex  $\rightarrow$  1 (random splines are uniform on  $[0,1]$ )



C: mean NP spline cumulative probability  $P(X \leq k)$



D:  $|\text{oracle mean } P(X \leq k) - \text{NP spline mean } P(X \leq k)|$

shrinking gap = oracle and NP spline converging in mean prediction

